

UNIT-5

Document Design and Graphics

A cluttered room with poorly designed lighting can make it difficult for people to find what they are looking for. Similarly, a cluttered document with poorly design elements can lead the readers to the same situation. The readers may be discouraged if they cannot find the information they need quickly and easily. Good technical writers must learn to design pages that are visually friendly to make readers to stay focused.

⊗. Designing the Document:

Document design is the process of choosing how to present all of the basic document elements so our document's message is clear and effective. When a document is well designed, readers understand the information more quickly and easily. Readers feel more positive about the topic and more accepting of its message.

Writers use design elements to keep readers interest. Design elements affect the page layout in writing document. These document design elements are as follows:

- 1) White Space: White space is space that is blank. It rests the eyes, separates chunks of information, and makes a document look inviting. Writers can create white space in margins, between paragraphs, between columns, around graphics etc.
- 2) Text: It refers to the words printed on the page or textual information on the screen. The readers read the text more easily and quickly if it is left-justified with ragged-right-edges. Highlighting features are print styles such as boldface, underline, italics, and capital letters that draw attention to words and phrases.
- 3) Headings: Headings are short titles that introduce the main idea of a selected portion of text. Like a formal outline, headings help our reader see the organization of document in one glance.

4) Graphics: Graphics are visual representations of information. They include many familiar visual aids such as tables, line graphs, pie graphs, and diagrams. Graphics are used in technical writing whenever information can be expressed better in a visual form than in words alone. For technical writers, a graphic is often used along with text to fully convey meaning.

5) Medium: In addition to considering white space, headings, and graphics, writers choose the best medium to use for their document. Possible choices include paper media or electronic media. Purpose and audience determine which medium to choose. In turn, the medium also influences the design.

⊗. Who Reads Graphics?

The audience, purpose and objectives are vital to any graphics.

Audience: Technical subjects such as engineering, marketing and medicine rely heavily on data that is presented visually. Therefore, readers of technical documents expect to see graphic aids in their reading. However, readers vary in their ability to understand graphics.

Generally, the more data in a graphic, the more difficult is to read. When deciding which graphic audience will best understand, we should ask ~~our~~ ourselves these questions about our reader(s):

- How much does my reader know about the subject?
- How interested is my reader in the subject?
- Do my readers include a technical audience?
- Will my audience be confused by technical information or figures?
- Does my audience's reading level tend to be higher or lower?

Purpose and Objectives: Audience is only one consideration in deciding what kind of graphic to use. We also must consider the purpose of our graphic as well as how much and what kind of information we want to convey.

To choose graphics that convey our meaning most effectively, we should ask ourselves these two questions: What is the purpose of the writing? How can graphics help to achieve that purpose? Then choose the graphic that best meets our needs.

⊗. Designing Graphics:

To help our readers interpret graphics quickly and easily, we should keep graphics simple and neat, integrate it with text, give credit for borrowed graphics, and use colour effectively.

- 1) Keep Graphics Neat and Simple: A neat and clean graphic is easy to read and interpret. We must leave enough space so that the graphics look uncluttered as well as make the graphics large enough so that the readers can visualize all parts clearly. Use number, decimal, percent symbol, etc properly in our document.
- 2) Integrate Graphics with Text: It is necessary to integrate graphics with the text. It means, both graphic and its description should be within a page. Use the word 'Table' to refer a formal table. In other cases, we can use the words figure to refer to any graphics, chart, diagram etc. Provide a title for every graphic. Both table title and figure title are often centered at the top and below the graphics respectively. The body part of our description should be in brief detail. We need to summarize the information to avoid the unnecessary details. The concluding part of the description should sum up the global trends shown on the graphic and make a comparison if possible.
- 3) Give credit for borrowed graphics: Give credit for a graphic if we do not compile it ourself or if we compile it using borrowed data. Place the word source below figure. Include the bibliographical reference for the source as we would a footnote or an endnote.
- 4) Use Color Effectively: Color often draws a reader's attention before the reader pays attention to the words. Color is a powerful design tool that can be used to:
 - Indicate a document's organization.
 - Emphasize or clarify an important point.
 - Support your text's meaning.
 - Make your document attractive.

⊗. How to use color effectively?

- 1) Avoid overusing color: More than three or four colors (including black and white as two colors.) can overwhelm a reader, and using one color (other than black) too often can be distracting.
- 2) Apply color consistently to elements throughout a document: For example, if we set major headings in red and set key words in black boldface, do not suddenly switch a major heading to blue or a key word to red boldface.
- 3) Remain sensitive to cultural identification with colors: International audiences associate colors differently from American audiences. For instance, Americans associate green with go (traffic lights) and red with danger (stop signs). However, green is a holy color to a Muslim and red is a sign of mourning to a South African.
- 4) Avoid unusual combination of colors: Some colors, such as purple on a blue background and orange on red, are difficult to read.

⊗. Constructing Graphics:

Before we draw lines and shapes, render text, or display and manipulate images, we need to construct graphics. The graphics represents drawing related to tables, charts and diagrams that are used to create graphical images. Different graphic constructions are presented as below:

1) Tables: Tables are used to present words and numbers that can be organized into categories of columns and rows. It is not a careless grouping of rows and columns but a technique of presenting data and variables.

Number: Each table should be numbered so that it may be easily identified.

Title: Every table should have clear and concise title.

Stub: It is the subcategories of variable, listed along the Y-axis.

Caption/Column headings: subcategories of variable, listed along X-axis.

Body: It contains cells for housing analyzed data.

Source: The source should be identified at the bottom of the table.

Footnotes: It is phrase or statement for further clarification of some specific items.

Example:

formal table means lines drawn with rules and consistent format.
 Informal without lines and rules inconsistent format.

Table: Distribution of Respondents by Gender

S.N.	Gender	Age Groups				Total	Percent
		21-30	31-40	41-50	51-60		
1.	Male	6	10	5	5	24	48.98
2.	Female	10	5	5	3	25	51.02
	Total	16	15	10	8	49	100

Source: Field Survey (2020).

2) Graphs/Charts: A graph is a chart that plots data using different types of statistical shapes that represent the total amount of observations in the data for that category. It is a visual aid that shows relationship among numerical data. Generally five types of graphs are used in graphic design such as bar graphs, pictographs, line graphs, Gantt charts, and pie charts.

a) Bar Graphs: The bar graphs represent data with rectangular bars. It presents data with rectangular bars with heights or lengths proportional to the value they represent. A bar graph can be either vertical or horizontal depending upon the choice of axis as base.

1) Simple Bar diagram: It is used to represent data involving only one variable classified on a spatial quantitative or temporal basis.

Example:

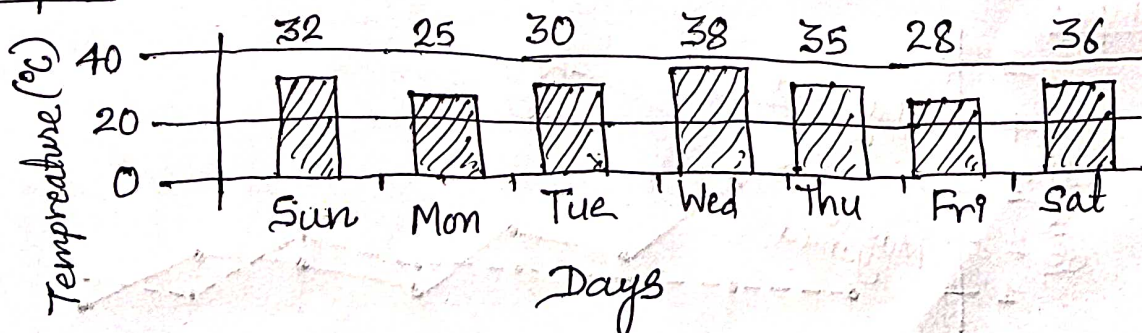


Fig: Temperature in Nepaljung.

11) Multiple Bar Diagram: It is used for comparing two or more sets of statistical data. Bars are constructed side by side to represent the set of values for comparison.

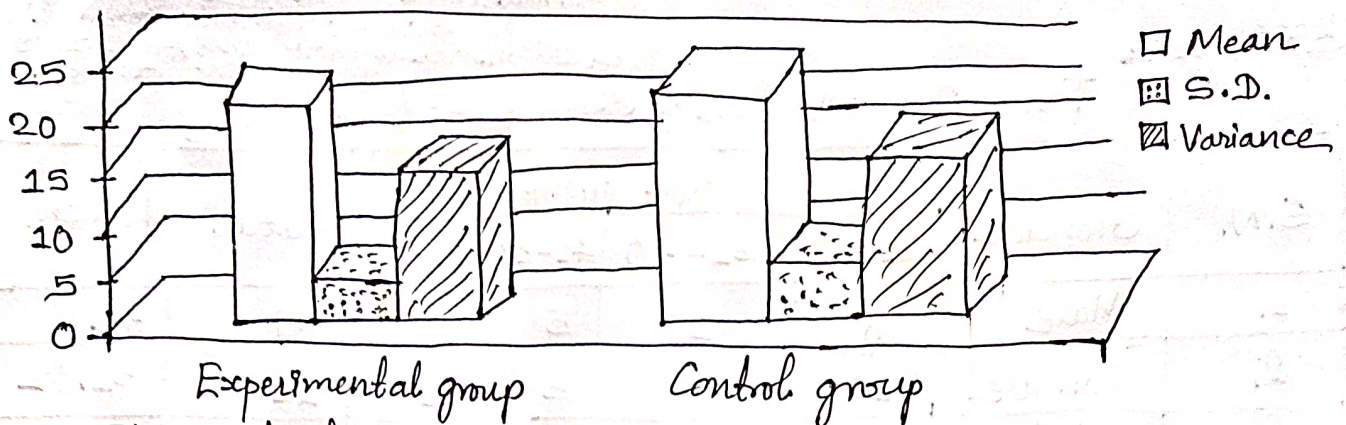


Fig: Pre-test Result of Experimental and Control Groups Students.

b) Line Graph: The line graph is similar to a bar graph in that it uses a horizontal axis and a vertical axis to compare numerical data. Instead of bars, however, this graph uses a line that depicts a trend.

Example:

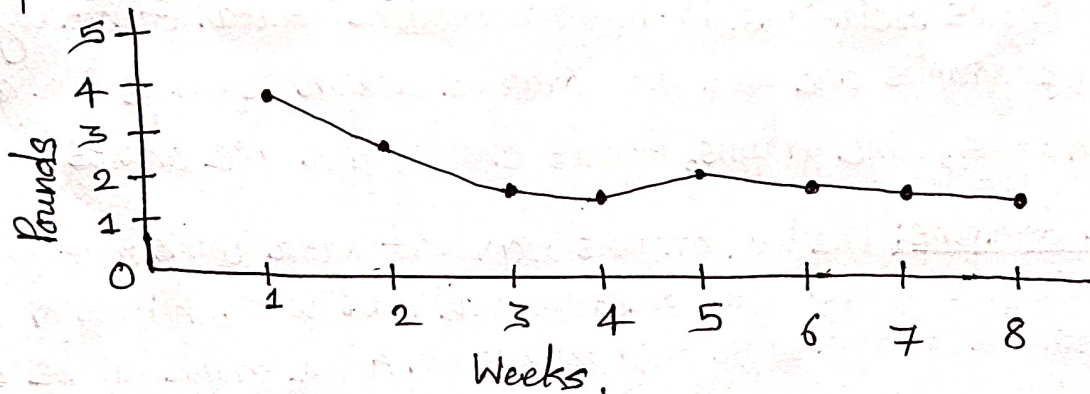


Fig: Fat loss (8 weeks).

Multiple line graph: Multiple line graph uses more than one line to compare data.

Example:

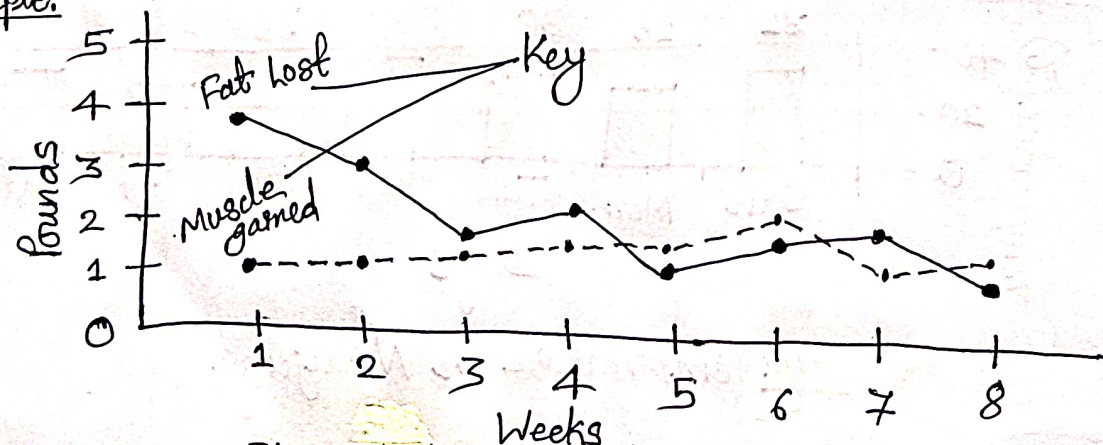


Fig: Fat loss (8 weeks)

c) Pictograph: A pictograph is a special kind of bar graph that uses pictures instead of bars to represent data. Like bar graphs, pictographs show differences in related data. In addition, pictographs add color and interest, are easy to read, and are especially appropriate for nonnative speakers of English. To be clear, the pictures or icons must be easily recognized and distinguishable from one another.

d) Pie Graph/Chart: A pie graph is a circular graphic that shows how parts relate to the whole. The whole totals 100 percent, with each piece of the pie representing a percentage of the whole. Notice that the pieces of the pie move clockwise from the twelve o'clock position from largest to smallest. A pie graph should contain no more than seven sections.

Example:

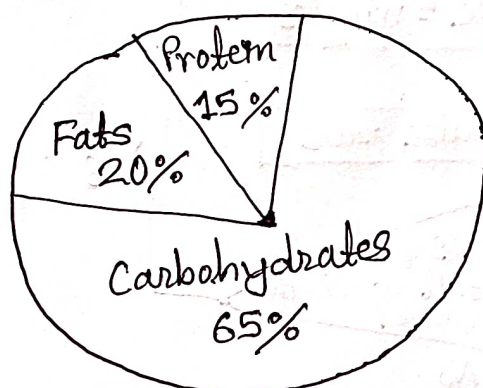


Fig: Daily Recommendations of Protein, Fat and Carbohydrate.

Gantt Chart: A Gantt Chart is a bar graph used to schedule the major tasks of a complex chart.

Example:

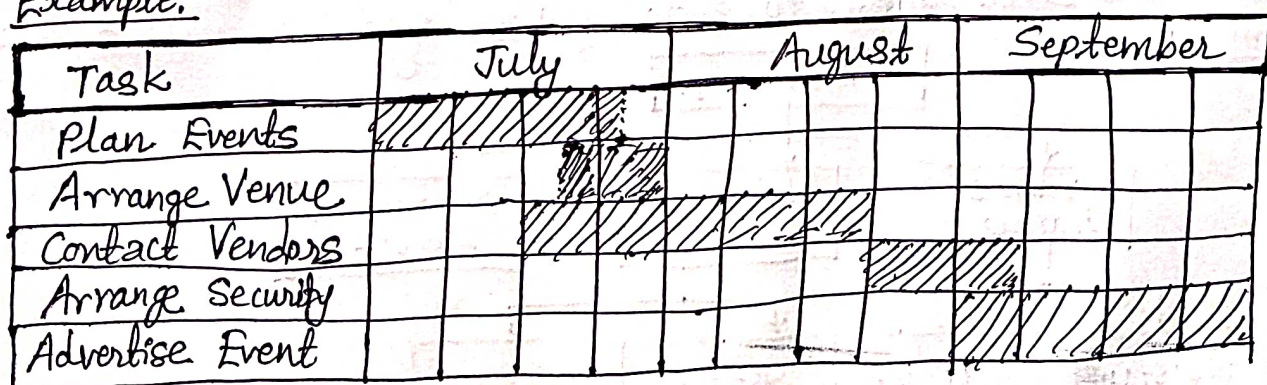


Fig: Planning Schedule for Health Fair.

Decision Flowchart: It is a special chart that uses symbols to indicate critical parts of making a decision. Ovals indicate where process starts and ends. Rectangles or squares indicate activities. Diamonds indicate places to make decisions. Arrows indicate the flow of the process.

Example:

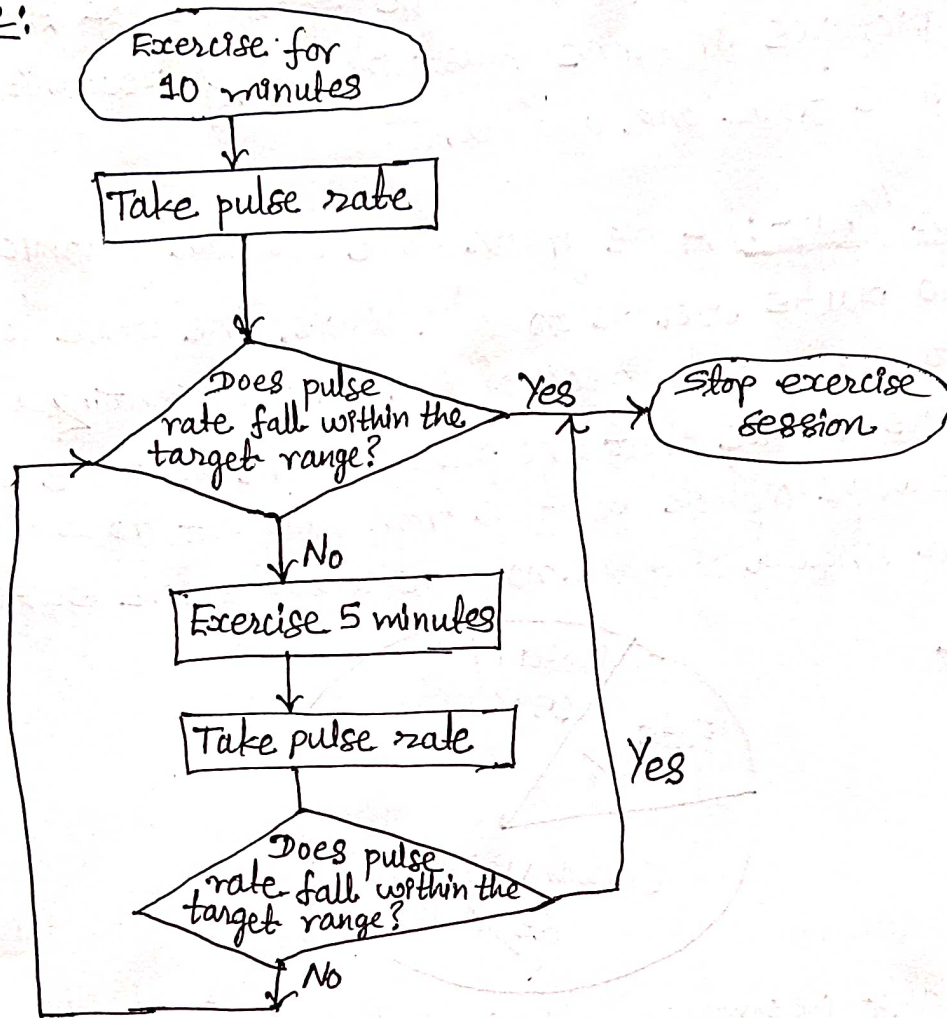


Fig: Process for Adjusting training rate.

Organizational Chart: An organizational chart is drawing with boxes, words and lines to show how an organization is structured.

Example:

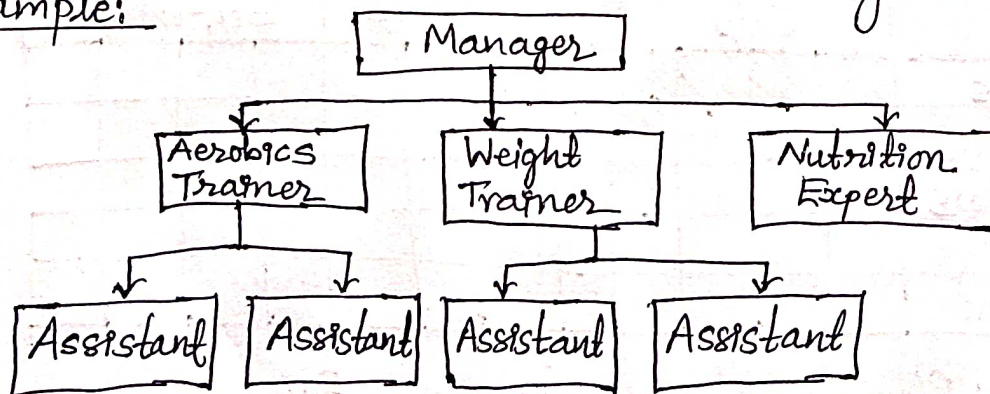


Fig: Model of organizational chart.